

Final Project Report

Title: Data-Driven Analysis of Rental Housing Stress in Regina and Saskatoon (2011 - 2023)



CS719: Data Science Project

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Abstract

Housing affordability has emerged as a pressing socio-economic challenge across Canadian cities, with growing concerns in mid-sized urban regions such as Regina and Saskatoon. This study presents a comprehensive, data-driven framework for quantifying rental housing stress by integrating multi-source datasets, including CMHC rental market statistics and Census-based income data, over the period 2011–2023. To capture the multi-dimensional nature of affordability, key indicators - rent-to-income ratio, single-person vulnerability, income inequality, and tax impact - are systematically engineered and normalized, and subsequently combined into a composite Housing Stress Score. A quantile-based classification approach is employed to categorize stress levels into low, medium, and high-risk segments, enabling consistent comparison across time and geography. The analysis reveals a sustained increase in rental stress, with a pronounced escalation post-2020. Significant disparities are observed across cities, zones, and housing types, with single-person households and larger units experiencing disproportionately higher financial pressure. Interactive visualization through Power BI dashboards enhances interpretability, uncovering spatial and temporal patterns in housing stress. The proposed framework not only improves understanding of rental affordability dynamics but also offers practical value for policymakers, urban planners, and residents by supporting targeted interventions and informed decision-making.

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1. Introduction

1.1. Main Idea

Housing affordability has emerged as a significant socio-economic challenge in modern urban environments. In recent years, rising housing costs, particularly in the rental sector, have placed increasing financial pressure on households across Canada. While major metropolitan areas such as Toronto and Vancouver often dominate discussions, mid-sized cities like Regina and Saskatoon are also experiencing growing affordability concerns that are less visible but equally important.

Rental housing plays a crucial role in providing flexibility and accessibility, especially for international students, young professionals, and lower-income groups. However, rent has been increasing at a faster pace compared to income, leading to a widening gap between housing costs and household earnings. This imbalance results in what is commonly referred to as rental housing stress, where a large share of income is spent on housing, leaving limited resources for other essential needs.

This study focuses on developing a structured and data-driven framework to analyze rental housing stress by integrating rent and income information. Instead of relying on a single indicator, the approach considers multiple aspects of affordability and represents them through a composite and interpretable Housing Stress Score. The framework is specifically designed to capture trends over time and variations across cities and zones, providing a more detailed understanding of housing stress in mid-sized urban settings.

1.2. Problem Statement

Despite the availability of detailed housing and income data, there is still a gap in how rental housing stress is measured and understood. Most existing analyses tend to look at individual indicators— such as average rent or median income—in isolation, without capturing how these factors interact in real-world situations.

This project addresses the problem through the following key challenges:

1. Growing gap between rent and income

Rental prices have been rising steadily, especially after 2020, while income growth has remained relatively slow. This imbalance has made housing less affordable over time and raises concerns about long-term financial sustainability for households.

2. Unequal impact across household types

Housing stress does not affect all households in the same way. Single-person households, for instance, often face higher financial pressure since they rely on a single income source. As a result, city-level averages can overlook these important differences.

3. Variation across zones within cities

Housing stress is not evenly distributed. Certain zones within a city experience significantly higher pressure due to factors like higher rents or lower availability of housing. Without zone-level analysis, these high-stress areas remain hidden.

4. Limitations of existing affordability measures

Common measures, such as rent-to-income ratio, provide only a partial view of affordability. There is a need for a more comprehensive and interpretable approach that combines multiple factors into a single measure for better comparison and decision-making.

1.3. Motivation

The motivation for this project comes from both practical needs and analytical interest. Housing affordability is a critical issue that directly affects financial stability and quality of life. When a large portion of income is spent on rent, households are left with limited capacity to manage other essential expenses, leading to long-term financial stress.

From a policy perspective, a clear and data-driven understanding of housing stress can support better decision-making. Identifying high-stress zones helps policymakers target interventions such as housing support programs, rental regulations, and infrastructure development more effectively.

Urban planners can also benefit from this analysis by understanding how housing stress varies across different zones. This enables more informed planning around housing supply, resource allocation, and balanced urban development.

From a data science standpoint, this project demonstrates how multiple datasets can be combined and transformed into meaningful insights. It applies techniques such as data integration, feature engineering, and composite score development to address a real-world problem.

Finally, the use of interactive dashboards makes the findings accessible and easy to interpret, allowing both technical and non-technical users to explore trends and make informed decisions.

1.4. Objective of the Study

The primary objective of this project is to develop a clear and interpretable framework for analyzing rental housing stress in Regina and Saskatoon. This is achieved through the construction of a composite Housing Stress Score along with a risk classification system.

The specific objectives of the study are as follows:

1. **Integrate rental and income data** across an extended time period (2011–2023), and estimate missing income values for non-census years using provincial income growth rates, ensuring a consistent and continuous dataset.
2. **Engineer meaningful features** that capture different dimensions of housing affordability, including rent-to-income ratio, single-person vulnerability, income inequality, and tax impact.
3. **Develop a composite Housing Stress Score** that represents the overall level of rental stress for each observation.
4. **Classify housing stress into categories** (Low, Medium, and High) using data-driven thresholds.

5. **Analyze trends over time**, with a focus on changes observed across the 13-year period (2011–2023).
6. **Compare housing stress across different dimensions**, including cities (Regina vs Saskatoon), zones, and bedroom types.
7. **Present the results through interactive dashboards**, providing clear and actionable insights.

1.5. Problem Specification

This project can be defined as an applied data science problem focused on socio-economic analysis, risk quantification, and the integration of multiple data sources.

Inputs

The analysis is based on a combination of rental and income datasets:

1. Rental market data

Information on average rent, vacancy rates, and apartment supply is obtained from the CMHC Rental Market Survey. This data is available at a detailed zone level for both Regina and Saskatoon.

2. Household income data

Median income values, including both before-tax and after-tax income, are obtained from the Census of Population for the years 2011, 2016, and 2021.

3. Estimated income for missing years

Since income data is not available for every year, the missing values are estimated using provincial income growth rates from Statistics Canada. These growth rates are applied to interpolate income values for intermediate years, ensuring continuity in the dataset.

By combining actual and estimated values, a consistent dataset is created covering the full period from 2011 to 2023, enabling a continuous time-series analysis of housing stress across both cities and their zones.

Outputs

The project produces the following key outputs:

1. A cleaned and integrated dataset combining rental and income data.
2. A set of engineered features representing key affordability indicators.
3. A Rental Housing Stress Score for each observation.
4. A risk classification (Low, Medium, High) based on the stress score.
5. A series of interactive dashboards that visualize trends, comparisons, and insights in a clear and accessible manner.

1.6. Organization of the Report

The remainder of this report is organized as follows:

Section 2 : reviews existing research and institutional work related to housing affordability and highlights key gaps that motivate this study.

Section 3 : presents the overall approach, including data sources, data integration, income estimation, feature engineering, normalization, and the development of the Housing Stress Score and risk classification framework.

Section 4 : discusses the results and analysis, including model validation and insights derived from visualizations and interactive dashboards.

Section 5 : concludes the report by summarizing the key findings and contributions of the study.

Section 6 : outlines suggestions for future work, including possible improvements and extensions to the proposed framework.

Section 7 : provides the list of references used in this study.

2. Background and Related Work

2.1. Existing Research and Institutional Work

Housing affordability in Canada has been extensively studied through institutional reports and official statistics. Organizations such as the Canada Mortgage and Housing Corporation (CMHC) and Statistics Canada provide detailed data and analysis on rental markets, income levels, and housing conditions across different regions.

CMHC regularly publishes rental market reports that include key indicators such as average rent, vacancy rates, and housing supply at a granular level. These reports are widely used to understand rental market dynamics and identify trends in housing demand and availability [1].

Similarly, Statistics Canada provides comprehensive datasets on household income, including both before-tax and after-tax income across different household types. In addition, reports such as *Housing affordability and core housing need* and publications from *The Daily* highlight broader affordability challenges, including the increasing financial burden on households and disparities across regions [2] [3].

Existing studies and reports primarily rely on indicators such as rent levels, income distribution, and affordability thresholds to assess housing conditions. While these sources provide valuable insights, they are often analyzed independently, focusing either on rental market conditions or income trends without combining both perspectives into a unified framework.

2.2. Common Analytical Approaches

Several analytical techniques have been used in prior studies to assess housing affordability:

- **Rent-to-Income Ratio (RTI):**

The most commonly used measure, representing the proportion of income spent on rent. It provides a basic indication of financial burden but lacks depth in capturing broader socio-economic conditions.

- **Composite Indices:**

Some studies use composite indices that combine multiple indicators (e.g., income, housing cost, and demographics) into a single measure. These approaches provide a more holistic view but often lack transparency in how weights are assigned.

- **Time-Series Analysis:**

Used to analyze trends in rent and income over time, helping identify periods of increasing affordability pressure.

- **Spatial Analysis:**

Geographic comparisons across regions or zones help identify high-risk areas where housing stress is concentrated.

These techniques form the foundation for analyzing housing affordability but are often applied independently rather than in an integrated framework.

2.3. Strengths and Limitations of Existing Approaches

Existing approaches provide valuable insights but also have several limitations:

Strengths:

- Simple indicators like RTI are easy to interpret and widely accepted
- Time-series analysis helps in understanding long-term trends
- Spatial analysis highlights regional disparities

Limitations:

- Most approaches rely on single indicators, missing the multidimensional nature of housing stress
- Lack of integration between income, household type, and geographic variation
- Limited ability to provide a comprehensive and interpretable risk measure
- Many composite models lack transparency in weighting methods

3. Approach

3.1. Overview

The goal of this study is to develop a clear way to analyze rental housing stress in Regina and Saskatoon by combining rent and income data into a single framework.

The process follows a step-by-step approach, starting with collecting data from different sources and preparing it for analysis. Since the datasets are not directly aligned, they are cleaned and integrated to create a consistent structure. After that, key indicators related to housing affordability are derived and brought to a common scale so they can be compared and combined.

These indicators are then used to compute a Housing Stress Score, which summarizes the overall level of rental pressure. Based on this score, observations are grouped into different risk categories to simplify interpretation.

The complete workflow is shown in Figure 1 on the next page, which outlines how the data moves from raw inputs to final outputs, including the dashboards used for visualization.

3.2. Data Sources

The analysis is based on publicly available datasets that reflect rental market conditions and household income across Regina and Saskatoon.

The main data sources used in this study are as follows:

- **CMHC Rental Market Survey [4]**

This dataset provides information on average rent, vacancy rates, and number of apartments at the zone level. It is used as the primary source to understand rental market trends.

- **Census of Population (2011, 2016, 2021) [5]**

These datasets provide median income values, including both before-tax and after-tax income for three groups: households, one-person households, and two-or-more-person households. This forms the base for analyzing income capacity.

Income data was only available for specific census years, it was not possible to

directly use it for all years.

- **Provincial Income Growth Rates (Statistics Canada) [\[6\]](#)**

These growth rates are used to estimate income for the missing years. Based on year-to-year growth, income values are extended to cover the full period from 2011 to 2023.

This combination of actual plus estimated income values helps in maintaining continuity in the dataset and allows better analysis of trends over time.

[These sources are widely recognized and used in policy-level analysis.]

3.3. Data Integration and Preparation

The datasets used in this study come from different sources and are not directly compatible in their original form. In particular, the rental data contains detailed information at the zone level, while the income data is available only at the city level and for census specific years. Because of this, several steps are required to prepare and integrate the data properly.

Methodology

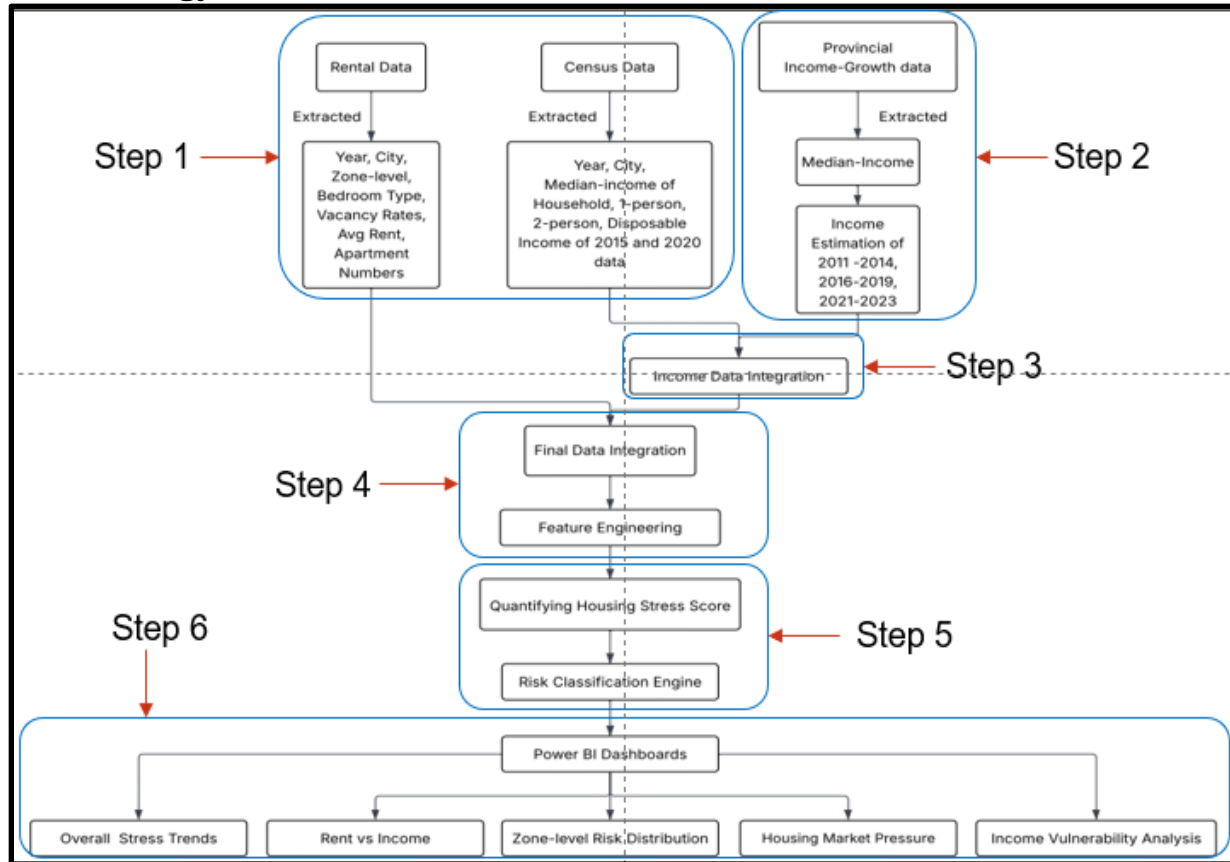


Figure 1: Methodology

Step 1 - The CMHC rental dataset includes fields such as city, zone, year, and bedroom type, which are used as the base structure for the analysis.

Step 2 - On the other hand, census income data is available only for the years 2011, 2016, and 2021, and only at the city level.

Therefore, the integration between rental and income data is carried out primarily using city and year, while zone-level variation is retained from the rental dataset.

From the census data, the following income variables were selected:

- Median before-tax income of households
- Median after-tax income of households
- Median after-tax income (1-person households)
- Median after-tax income (2+ person households)

Step 3 - Since these values are not available for every year, income needs to be estimated Statistics Canada, specifically using the median income (excluding zeros) as a reference.

Step 3.1 - year-to-year growth rates are calculated using:

$$Growth\ Rate_t = \frac{Income_t - Income_{t-1}}{Income_{t-1}}$$

These growth rates are in percentage form, then used to estimate missing income values.

Step 3.2 - For forward estimation (future years), the formula is:

$$Income_t = Income_{t-1} \times (1 + Growth\ Rate_t)$$

For backward estimation (earlier years), the formula is:

$$Income_t = \frac{Income_{t+1}}{1 + Growth\ Rate_{t+1}}$$

Using these two approaches, income values are estimated for all missing years between 2011 and 2023. The same process is applied consistently across all income categories.

(t = Current Year)

($t - 1$ = Previous Year)

($t + 1$ = Next Year)

For example, let us calculate after-tax income for the years 2016, and 2014

First, Calculate the growth rate: for the year 2016, and 2015.

(To calculate growth rate refer step 3.1)

$$Growth\ Rate_{2015} = \frac{47300 - 47000}{47000} = 0.64 \%$$

$$Growth\ Rate_{2016} = \frac{46200 - 47300}{47300} = -2.33 \%$$

Now, use above growth rate percentage, in estimating final income for the respective years.

(To calculate current year and previous year income refer- step 3.2 for the formulation)

$$\begin{aligned} \text{Income}_{2016} &= 70299 \times (1 + (-0.0233)) = 68,593 \\ \text{Income}_{2014} &= \frac{70299}{1 + 0.0064} = 69,782 \end{aligned}$$

This methodology is used to compute income values for all 3 categories for the remaining years (2011 – 2014, 2016-2019, and 2021-2023)

- Median after-tax income of households
- Median after-tax income (1-person households)
- Median after-tax income (2+ person households)

Step 3.3 - In addition to after-tax income, before-tax income is also required.

Since it is not available for all years, a tax ratio approach is used.

Step 3.3.1 - tax ratios are calculated for known years (2011, 2015, 2020) using:

$$TR = \frac{\text{After-Tax Income}}{\text{Before-Tax Income}} \quad (TR = \text{Tax Ratio})$$

Since both values were available from census data, tax ratios can be directly calculated for those years.

Step 3.3.2 – Tax Ratios for the missing years were estimated using interpolation.

This is done by first calculating the difference between two known tax ratios.

For example, between 2015 and 2020:

$$\Delta TR = TR_{2020} - TR_{2015}$$

This difference is then divided by the 5-year gap to obtain the annual change:

$$\text{Annual Change} = \frac{\Delta TR}{5}$$

For Example:

$$\Delta TR = 0.011$$

(5 = year gap)

$$Annual\ Change = \frac{0.011}{5} = 0.0022$$

This annual change is then used to estimate tax ratios for intermediate years:

$$TR_t = TR_{base\ year} + ((t - base\ year) \div 5) \times Annual\ Change$$

For Example:

$$\begin{aligned} TR_{2016} &= 0.858 + ((2016 - 2015) \div 5) \times 0.011 \\ &= 0.858 + (1 \div 5) \times 0.011 \\ &= 0.858 + 0.0022 \end{aligned}$$

$$TR_{2016} = 0.8602$$

base year = 2015

$$TR_{2015} = 0.858$$

Step 3.3.3 - Once the tax ratio for a year is obtained, before-tax income is calculated as:

$$Before - Tax\ Income_t = \frac{After - Tax\ Income_t}{TR_t}$$

$$Before - Tax\ Income_{2016} = \frac{68,593}{0.8602} = 79,740$$

This method is applied consistently to estimate before-tax income for all required years.

Finally, the rental data and the processed income data are combined based on city and year. Additional transformations, such as converting average monthly rent into annual rent by 12:

$$Annual\ Rent = Avg\ Rent \times 12$$

are applied to ensure consistency in units across all variables.

Step 4 - After these steps, the final dataset is complete, consistent across all years from 2011 to 2023, and ready for further analysis.

Section 3.4 – provides in detail explanation of the feature engineering and normalization part

Step 5 – Section 3.5 mentions the housing stress score logic, followed by Risk classification engine in the Section 3.6

Step 6 – Finalize clean data, with rental housing stress score and risk classification results will be deployed in Microsoft Power BI (Desktop Version) for preparing interactive dashboards.

3.4. Feature Engineering and Normalization

After preparing the dataset, a set of derived variables is created to capture different dimensions of housing affordability. These features are based on rent and income values and are later used in the housing stress model.

Feature 1: Rent-to-Income Ratio (RTI)

Meaning: Measures the share of income spent on rent:

$$RTI = \frac{\text{Annual Rent}}{\text{Median After – Tax Income (Households)}}$$

Feature 2: Single Person Vulnerability (SPV)

Meaning: To capture vulnerability at an individual level, a separate measure is created for single-person households

$$SPV = \frac{\text{Annual Rent}}{\text{Median After – Tax Income (1 Person)}}$$

Feature 3: Income Inequality (II)

Meaning: represent the difference in earning capacity between household types

$$II = \text{Median After – Tax Income (2 + Person)} - \text{Median After – Tax Income (1 Person)}$$

Feature 4: Tax Impact (TI)

Meaning: indicates the portion of income that is not available for spending and indirectly affects affordability.

$$TI = (\text{Median Before – Tax Income}) - (\text{Median After – Tax Income})$$

Together, these features represent different aspects of housing stress, including rent pressure, income availability, and household-level differences.

Since these features are measured in different units and ranges, they are normalized before being used in the model. A **min-max normalization** approach is applied to scale all features between 0 and 1:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

This ensures that no single feature dominates the model due to scale differences and allows all variables to contribute comparably.

These normalized features are then used in the next step to construct the housing stress model.

3.5. Housing Stress Score Model

After deriving and normalizing the key features, a composite Housing Stress Score is constructed to represent the overall level of rental pressure for each observation.

The idea behind this model is to combine multiple aspects of affordability into a single value, instead of relying on any one indicator. Each feature captures a different dimension, such as rent burden, individual vulnerability, income differences, and effective income after tax.

The **Housing Stress Score** is calculated as a weighted combination of the normalized features:

$$HSS = 0.40 \cdot RTI_{norm} + 0.30 \cdot SPV_{norm} + 0.15 \cdot II_{norm} + 0.15 \cdot TI_{norm}$$

where:

- RTI_{norm} = normalized Rent-To-Income Ratio
- SPV_{norm} = normalized Single-Person Vulnerability
- II_{norm} = normalized Income Inequality Indicator
- TI_{norm} = normalized tax impact indicator

3.6. Risk Classification

After calculating the Housing Stress Score, the results are categorized into different risk levels to make interpretation easier.

In this study, a **quantile-based approach** is used to divide the data into three categories:

- Low Risk
- Medium Risk
- High Risk

The thresholds are determined based on the distribution of the Housing Stress Score. Specifically, the lower and upper cut-off values are taken from the data itself, so that the observations are reasonably distributed across the three categories.

For this dataset, the approximate threshold values are:

- **Low Stress:** $\text{Score} \leq 0.37$
- **Medium Stress:** $0.37 < \text{Score} \leq 0.52$
- **High Stress:** $\text{Score} > 0.52$

This approach ensures that the classification adapts to the underlying data and reflects relative stress levels rather than relying on predefined assumptions.

The categorized results are then used in the dashboard visualizations to highlight areas and groups experiencing higher levels of housing stress.

3.7. Example Illustration

To demonstrate how the model works, a simple example is considered using sample values.

Suppose for a given observation:

- Average Monthly Rent = \$1,100
- Annual Rent = \$13,200
- Median After-Tax Income (Households) = \$70,000
- Median After-Tax Income (1 Person) = \$40,000
- Median After-Tax Income (2+ Person) = \$85,000
- Median Before-Tax Income = \$95,000

Step 1: Calculate Engineered Features

$$RTI = \frac{13,200}{70,000} \approx 0.189$$

$$SPV = \frac{13,200}{40,000} \approx 0.33$$

$$II = 85,000 - 40,000 = 45,000$$

$$TI = 95,000 - 70,000 = 25,000$$

Step 2: Normalize Features

(Assuming min-max scaling is already applied across the dataset)

Suppose the normalized values are:

- $RTI_{norm} = 0.60$
- $SPV_{norm} = 0.55$
- $II_{norm} = 0.50$
- $TI_{norm} = 0.45$

Step 3: Calculate Housing Stress Score

$$HSS = 0.40 (0.60) + 0.30 (0.55) + 0.15 (0.50) + 0.15 (0.45)$$

$$HSS = 0.24 + 0.165 + 0.075 + 0.0675$$

$$HSS = 0.5475$$

Step 4: Assign Risk Category

Based on the classification thresholds:

- Low Risk: $HSS \leq 0.37$
- Medium Risk: $0.37 < HSS \leq 0.52$
- High Risk: $HSS > 0.52$

Since:

$$HSS = 0.5475 > 0.52$$

The Observation is classified as: **High Risk**

This example shows how raw input values are transformed step-by-step into a final Housing Stress Score and corresponding risk category. It also highlights how different factors contribute to the overall stress level.

3.8. Implementation Tools

The analysis in this study is carried out using a combination of standard data processing and visualization tools.

1. **Microsoft Excel:** for initial data handling and organizing intermediate calculations.
2. **Jupyter Python Notebook:** for income and complete data integration, data preprocessing, feature engineering, normalization, and model computation.
3. **Microsoft Power BI:** used to develop interactive dashboards for visualizing the results and identifying patterns across different dimensions.

3.9. Advantages and Limitations

Advantages

1. The Model combines multiple factors such as rent, income, and household differences into a single framework, which gives a more complete view of housing stress compared to using only one indicator.
2. The use of estimated income values allows the dataset to cover a continuous period from 2011 to 2023, making it possible to analyze trends over time.
3. The model remains simple and interpretable, as each component and its contribution can be clearly understood.

Limitations

1. The income estimation is based on provincial growth rates, which assumes that income trends at the provincial level apply equally to specific cities. This may not fully capture local variations.
2. The weights used in the model are based on logical reasoning and prior research, rather than statistical optimization, which may affect the precision of the final score.
3. The model does not include other relevant factors such as demographic characteristics or policy changes, which could also influence housing affordability.
4. Dynamic threshold approach could improve adaptability.

4. Results and Analysis

4.1. Model Validation and Future Insights

1. Feature Correlation with Housing Stress Score

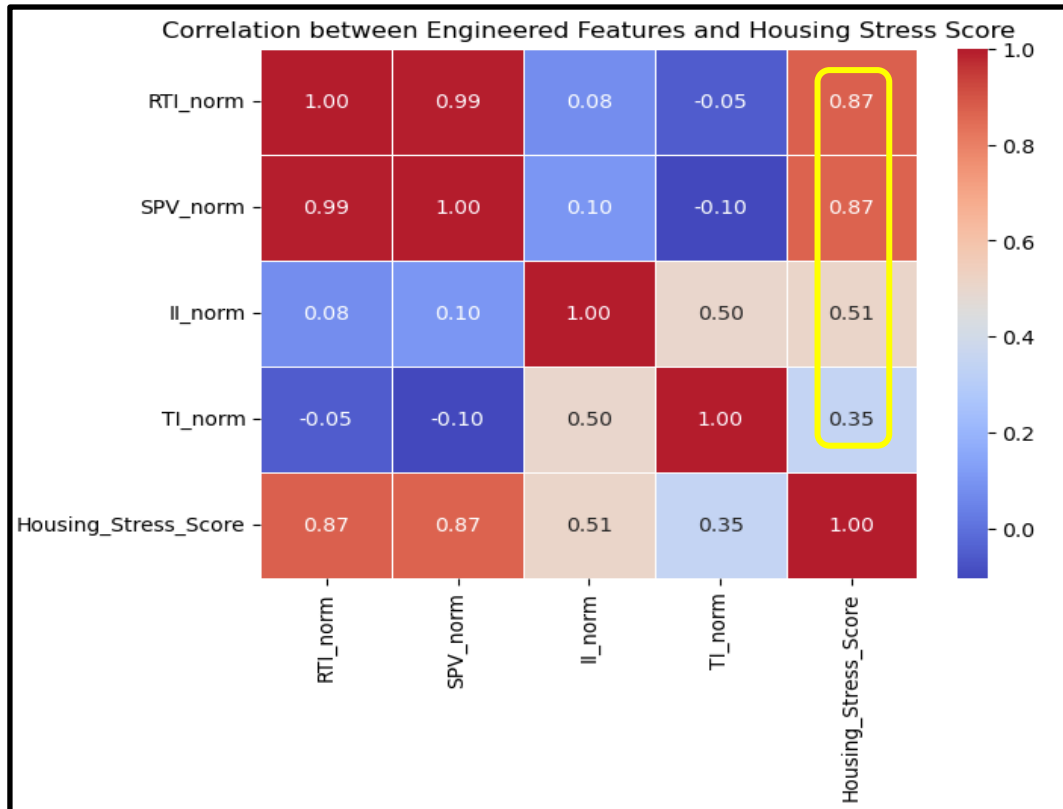


Figure 2: Feature Correlation

Interpretation:

- **RTI_norm vs Housing Stress Score (0.87) – Strong Positive Correlation**
 - As rent burden (Rent -to Income Ratio) increases, the housing stress score also increases.
- **SPV_norm vs Housing Stress Score (0.87) – Strong Positive Correlation**
 - Areas where single-person households face higher rent pressure tend to have higher housing stress.
- **II_norm vs Housing Stress Score (0.51) – Moderate Positive Correlation**
 - When the income gap between single-person and multi-person households increases, rental stress tends to increase.

- **TI_norm vs Housing Stress Score (0.35) – Weak Positive Correlation**
 - Higher tax burden (less disposable income) increases rental stress.

2. Feature Contribution to Housing Stress Score

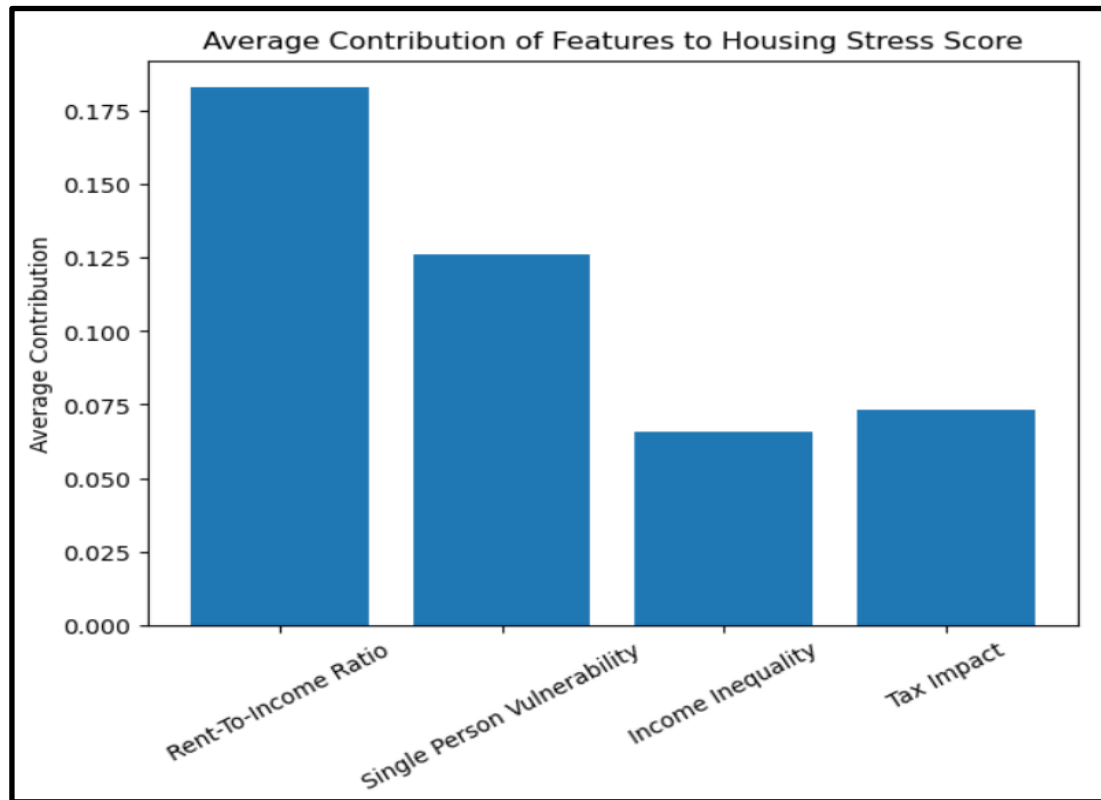


Figure 3 - Feature Contribution

Interpretation:

- **RTI (Rent-to-Income Ratio) – Highest Contribution (~0.18)**
 - Rent burden is the most dominant factor in determining housing stress.
 - Higher rent relative to income directly increases overall stress levels.
- **SPV (Single Person Vulnerability) – Moderate to High Contribution (~0.13)**
 - Single-person households contribute significantly to housing stress.
 - This indicates that financially exposed groups face greater affordability challenges.

- **TI (Tax Impact) – Lower Contribution (~0.07)**
 - Tax burden has a smaller but noticeable influence on housing stress.
 - Reduced disposable income slightly increases stress levels.
- **II (Income Inequality) – Lowest Contribution (~0.06)**
 - Income differences between household types have the least direct impact.
 - While relevant, it plays a secondary role compared to affordability pressure.

3. Distribution of Housing Stress Risk Categories (%)

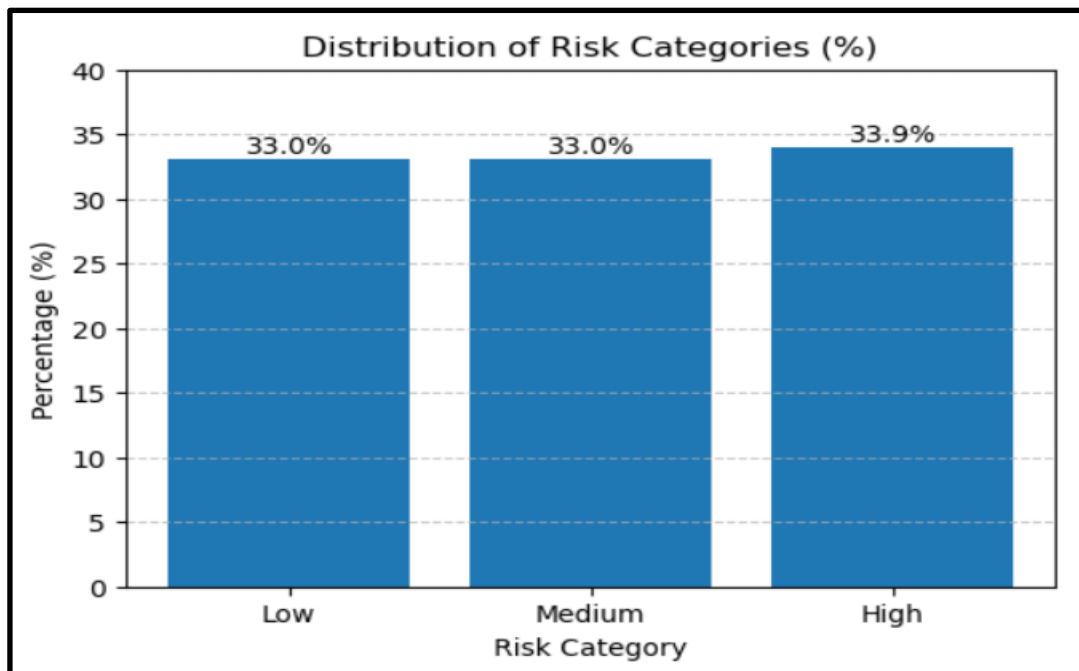


Figure 4 – Risk Distribution

Interpretation:

- **Balanced distribution across all categories (~33% each)**
 - Low, Medium, and High risk categories are almost equally distributed.
 - This indicates that the classification is not biased towards any single category.
- **Effective threshold selection**
 - The thresholds used for classification successfully divide the data into meaningful groups.
 - Each category captures a significant portion of observations.

- **Presence of all risk levels in the dataset**
 - The dataset includes a mix of low, moderate, and high stress conditions.
 - This supports a comprehensive analysis of housing stress.
- **Model output is well-structured for analysis**
 - The balanced categories allow clear comparison across cities, zones, and time periods.

The balanced categories allow clear comparison across cities, zones, and time periods.

Overall, model validity is supported through strong feature correlations and a balanced distribution of risk categories, indicating that the framework produces consistent and meaningful results.

4.2. Dashboards

1. Overall Rental Stress

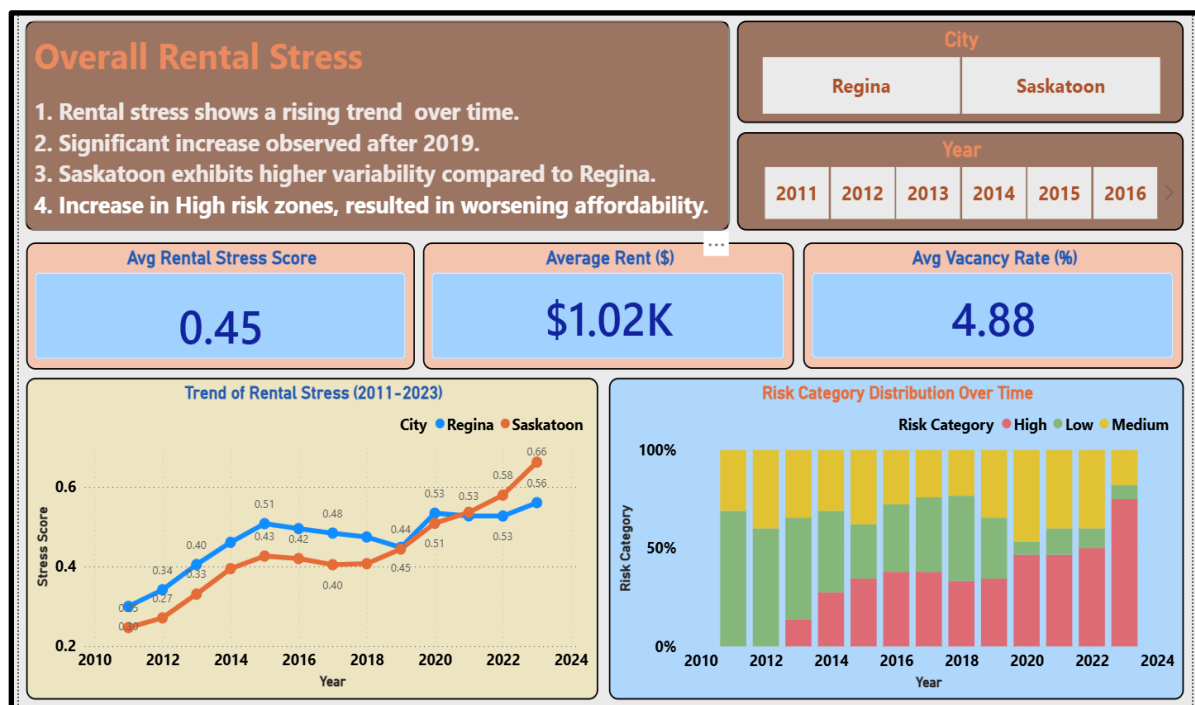


Figure 5 - Overall Rental Stress

This dashboard gives a quick overview of how rental stress has changed over time across Regina and Saskatoon.

- **KPIs**

At a summary level, the average housing stress score is around 0.45, which already indicates a moderate level of stress. At the same time, average rent is about \$1,020, while vacancy rates remain relatively low, suggesting some pressure on housing supply.

- **Trend of Rental Stress**

Looking at the trend line more closely, both cities show an upward movement, but the pattern differs. Regina shows more noticeable fluctuations over the years, indicating that its rental market experiences more short-term volatility. In contrast, Saskatoon follows a relatively smoother trend, but with a much stronger overall increase. For example, Saskatoon's stress score rises from around 0.25 in 2011 to about 0.66 in 2023, whereas Regina increases from approximately 0.30 to 0.56 over the same period.

This suggests that while Regina experiences more year-to-year variation, Saskatoon has seen a larger long-term increase in housing stress, indicating a more significant overall shift in affordability.

- **Risk Category Distribution over time**

The risk distribution chart adds another important layer. In earlier years, most zones fall under low or medium risk. However, as we move toward recent years, especially after 2019, there is a visible shift toward high-risk categories. This means that more areas are now experiencing higher levels of housing stress.

Overall, the dashboard clearly shows that rental stress is not only increasing over time but is also spreading across more zones. This indicates a gradual decline in rental housing affordability, with stronger pressure observed in the post 2020 years.

2. Rent vs Income Dynamics

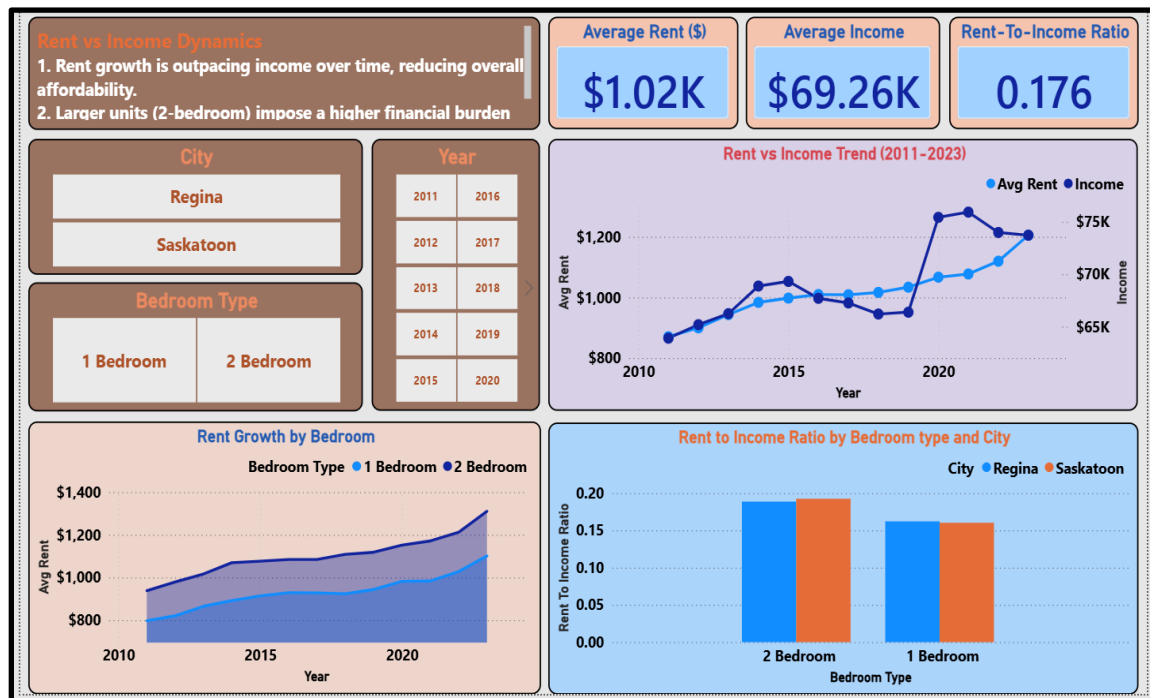


Figure 6 - Rent vs Income Dynamics

This dashboard focuses on one of the most important questions in housing affordability - whether income growth is keeping up with rising rent.

- KPIs**

At a summary level, the average rent is around \$1,020, while the average household income is approximately \$69,260. This results in a rent-to-income ratio of about 0.176, meaning households are spending close to 18% of their income on rent. While this may appear manageable at a glance, the trend over time tells a more concerning story.

- Rent vs Income trend**

Looking at the trend, both rent and income increase over the years, but they do not grow at the same pace. In the earlier years, households were spending roughly 15–16% of their annual income on rent. As we move toward 2023, this increases to nearly 20%. This steady rise clearly indicates that affordability is getting worse, with a larger portion of income being consumed by rent.

- **Rent Growth by Bedroom**

The bedroom-wise analysis provides another important insight. The area chart shows the difference between 1-bedroom and 2-bedroom rents, and this gap continues to increase over time. Initially, the difference is around \$150–\$200, but it gradually widens and reaches approximately \$400–\$500 in recent years. This clearly indicates that 2-bedroom units are becoming significantly more expensive compared to 1-bedroom units.

- **Rent-to-income rate by Bedroom type and city**

This pattern is also reflected in the rent-to-income ratio. Two-bedroom units consistently have higher ratios, meaning households living in larger units are spending a greater share of their income on rent.

Overall, the dashboard highlights a key issue: rent is rising faster than income, and the gap between different housing types is also increasing. Together, these factors are putting increasing financial pressure on households, especially to households requiring larger living spaces.

3. Rent Difference Between 1-Bedroom and 2-Bedroom Units by Zone and City

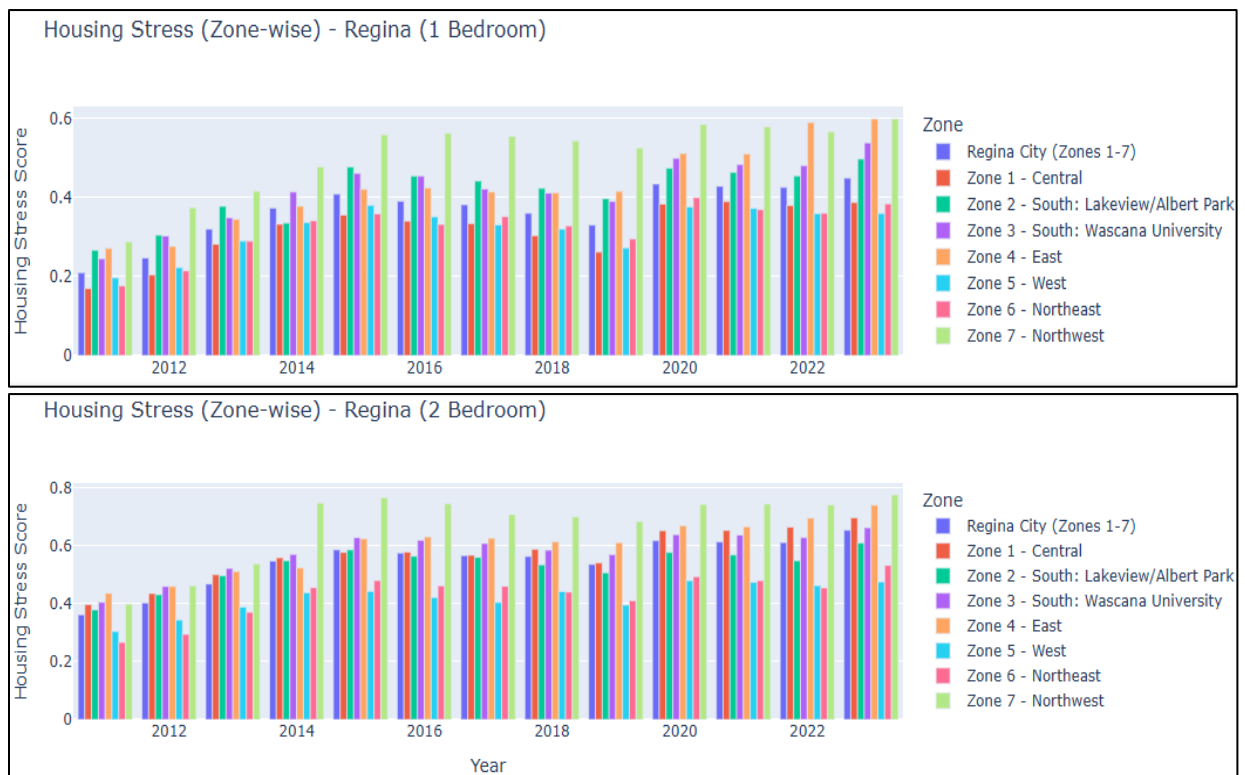


Figure 7 - Rent Difference by Bedroom Type-Regina

Figure 7

Rent Difference Analysis – Regina (1-Bedroom vs 2-Bedroom)

- This visualization focuses on how the cost difference between 1-bedroom and 2-bedroom apartments behaves across different zones in Regina.
- Across all zones in Regina, 2-bedroom units are consistently more expensive than 1-bedroom units, and this gap gradually increases over time. This indicates growing financial pressure on households requiring larger housing.
- Both bedroom types follow a similar trend—steady growth from 2011 to around 2016, slight stabilization between 2017 and 2019, and a sharper increase after 2020. Stress levels reach their highest values by 2023.
- At the zone level, differences exist but remain relatively balanced. Zones like Zone 7 and Zone 4 show higher stress, while Zone 5 and Zone 6 remain comparatively lower.

However, the gap between 1-bedroom and 2-bedroom units stays fairly consistent across zones.

- Overall, Regina shows a gradual and stable increase in housing stress, with larger units consistently less affordable.

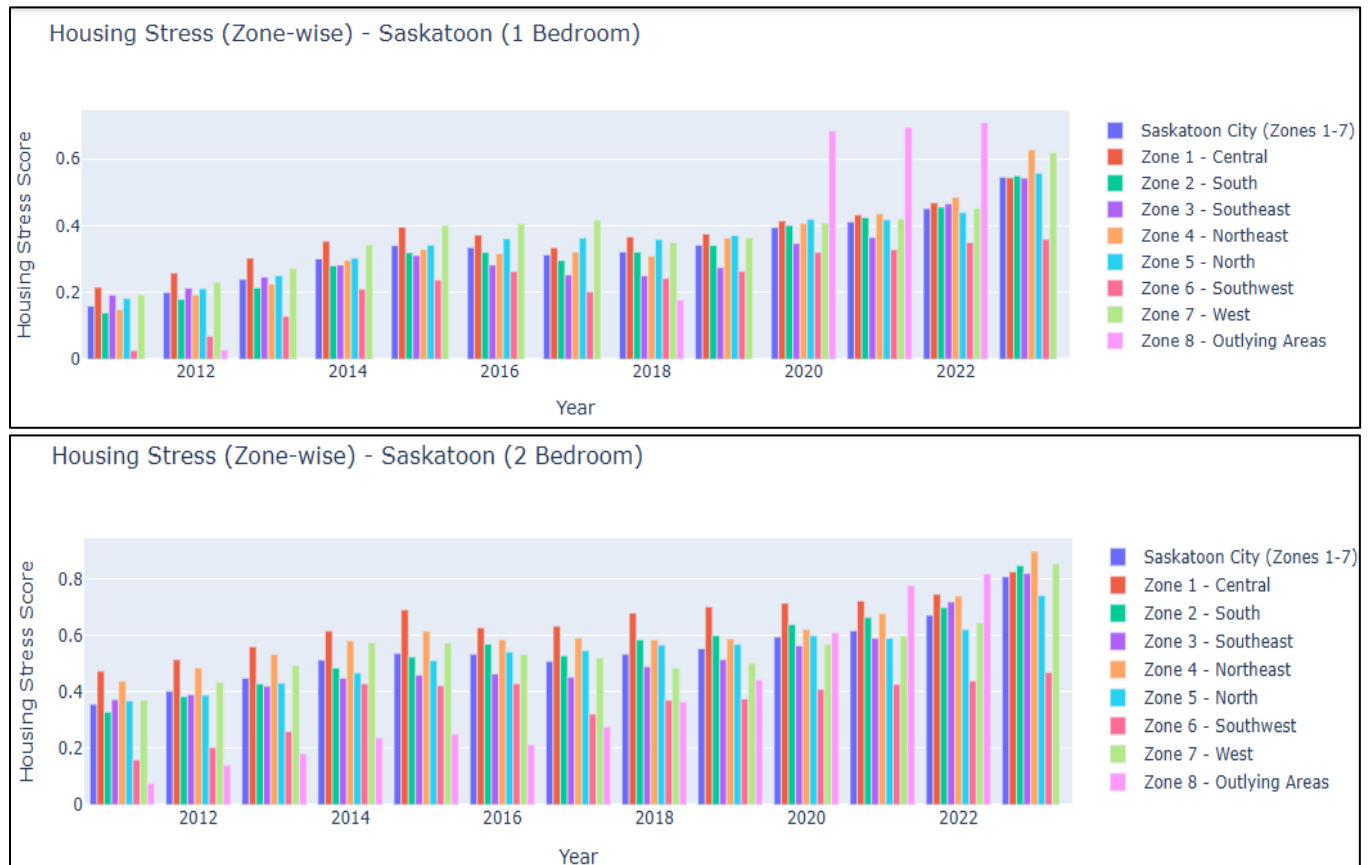


Figure 8 - Rent Difference by Bedroom Type-Saskatoon

Rent Difference Analysis – Saskatoon (1-Bedroom vs 2-Bedroom)

- In Saskatoon, the difference between 1-bedroom and 2-bedroom units is more pronounced compared to Regina. 2-bedroom units consistently show much higher stress levels, and the gap between the two types increases significantly over time.
- Both bedroom types show gradual growth in the early years, but after 2020, there is a sharp rise, especially for 2-bedroom units. This indicates a stronger increase in affordability pressure in recent years.

- A key observation is the behavior of Zone 8 (Outlying Areas), which shows sudden spikes post-2020, particularly in 2-bedroom units. This suggests that outer areas are also becoming increasingly expensive, not just central zones.
- Overall, Saskatoon shows a more aggressive increase in housing stress, with a wider and growing gap between 1-bedroom and 2-bedroom units, making larger housing significantly less affordable.

Overall Comparison: Regina vs Saskatoon

Both cities show a similar overall pattern—housing stress increases over time, with a noticeable rise after 2020. However, the intensity and behavior of this increase differ between the two.

Regina shows a more stable and gradual trend. While stress levels do increase, the growth is relatively controlled, and the gap between 1-bedroom and 2-bedroom units remains consistent across zones. This suggests a more balanced rental structure.

In contrast, Saskatoon experiences a sharper and more uneven increase in housing stress. The gap between 1-bedroom and 2-bedroom units is wider and continues to grow, especially in recent years. Certain zones, such as Zone 8, show sudden spikes, indicating higher variability and localized affordability pressure.

Overall, Regina reflects steady worsening affordability, whereas Saskatoon shows a faster and more pronounced decline in affordability, particularly for larger housing units.

4. Vulnerable Group Analysis

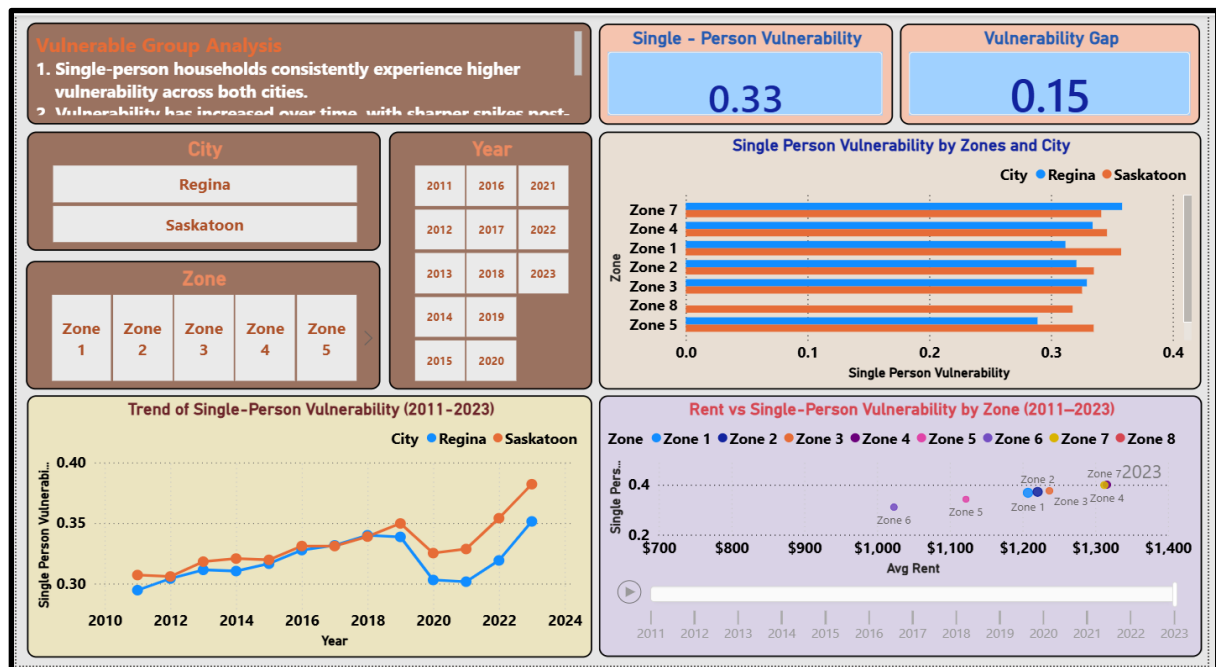


Figure 9 - Vulnerable Group Analysis

This dashboard focuses on identifying the most vulnerable group in the rental market — particularly comparing single-person households across cities and zones.

- **KPIs**

The Single Person Vulnerability (SPV) score is around 0.33, which means single individuals are spending nearly 33% of their income on rent. When compared with the overall rent-to-income ratio (~0.18), the gap is about 15 percentage points, indicating that single-person households face significantly higher financial pressure.

- **Single Person Vulnerability by Zones and City**

At the zone level, clear differences are visible across both cities. In Regina, Zone 7 (Northwest) shows the highest vulnerability, followed by Zone 4 (East) and Zone 3 (South – Wascana University). Lower vulnerability is seen in Zone 6 (Northeast) and Zone 5 (West).

In Saskatoon, Zone 1 (Central) has the highest vulnerability, followed by Zone 4 (Northeast) and Zone 7 (West), indicating strong pressure in central and key residential areas.

- **Trend of Single-Person Vulnerability**

Looking at the trend over time, both cities show a steady increase in vulnerability from 2011 to 2019, followed by a noticeable dip around 2020, and then a sharp increase again in recent years. This suggests that while there was a temporary relief, affordability pressure has intensified again after 2020.

- **Rent vs Single-Person Vulnerability by Zone**

The rent vs SPV analysis further strengthens this observation. In the early years, most zones lie in the bottom-left region, indicating lower rent and lower vulnerability. Over time, these points gradually move toward the top-right, especially after 2020. This shows a strong positive relationship—as rent increases, single-person vulnerability also increases.

Overall, the analysis clearly shows that single-person households are more exposed to rental stress, and this vulnerability has increased significantly over time, especially in high-rent zones.

5. Zonal-Level Risk Distribution



Figure 10 - Zonal-Level Risk Distribution

This dashboard drills down from city-level insights to zone-level risk analysis. It helps identify which specific areas are consistently under higher housing stress.

- Zone-Level Rental Stress Ranking**

Looking at the zone-level ranking, a similar pattern to the SPV analysis is observed. In Regina, Zone 7 (Northwest) and Zone 4 (East) consistently show the highest stress levels, while zones like Zone 5 and Zone 6 experience comparatively lower stress. In Saskatoon, Zone 1 (Central) and Zone 4 stand out as the more stressed zones, indicating higher affordability pressure in these areas.

- Change in Rental Stress by Zone**

This chart highlights how conditions have evolved over time. The dominance of the orange bars (Saskatoon) indicates that several zones in Saskatoon have experienced a larger increase in stress compared to Regina. This aligns with earlier observations where Saskatoon showed a sharper rise in overall housing stress.

- **Zone-Level Stress Evolution**

The heatmap of stress evolution provides a detailed view of how each zone has progressed from 2011 to 2023. It clearly shows a gradual shift from lower to higher stress levels across most zones, with more intense values appearing in recent years, especially after 2020.

- **Risk Composition by Zone**

The risk composition chart breaks down how zones are distributed across Low, Medium, and High risk categories. Some zones have a larger share under high-risk conditions, while others still retain a balance between medium and low risk. This highlights that housing stress is not evenly distributed and varies significantly by location.

Overall, this dashboard reinforces earlier findings by showing that high-stress zones remain consistent across analyses, while also highlighting that stress levels have increased over time and are more concentrated in specific areas, particularly in Saskatoon.

6. Geospatial Risk & Affordability Analysis

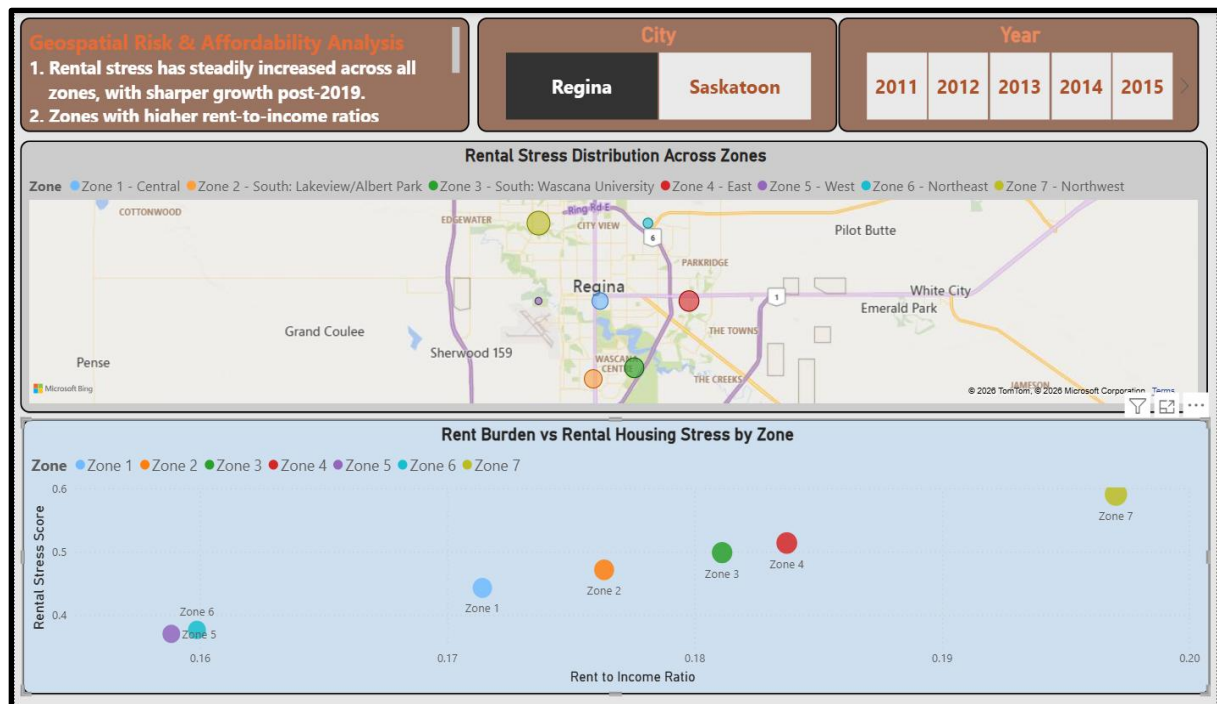


Figure 11 - Geospatial Risk & Affordability Analysis

This dashboard combines spatial and financial analysis to show where rental stress is concentrated and how it relates to affordability across zones.

- **Rental Stress Distribution Across Zones**

In this visual, each zone is represented using bubbles, where the size of the bubble indicates the level of housing stress. Larger bubbles clearly highlight zones experiencing higher stress. For example, in Regina, Zone 7 stands out with a larger bubble, indicating higher rental pressure compared to other zones.

- **Rent Burden vs Rental Housing Stress by Zone**

The rent burden vs housing stress scatter plot further explains this pattern by showing the relationship between rent-to-income ratio and housing stress score. A clear positive relationship is visible—as the rent-to-income ratio increases, the housing stress score also increases.

Zones positioned toward the top-right corner represent the most critical cases. For instance, Zone 7 in Regina appears in this region, indicating both a high rent burden and

high stress level. In contrast, zones located in the bottom-left area experience relatively lower rent pressure and lower stress.

Overall, this dashboard confirms that housing stress is not randomly distributed—it is strongly linked to rent burden and varies across locations. Zones with higher rent-to-income ratios consistently experience higher stress, reinforcing the core finding of this study.

7. Rental Market Pressure

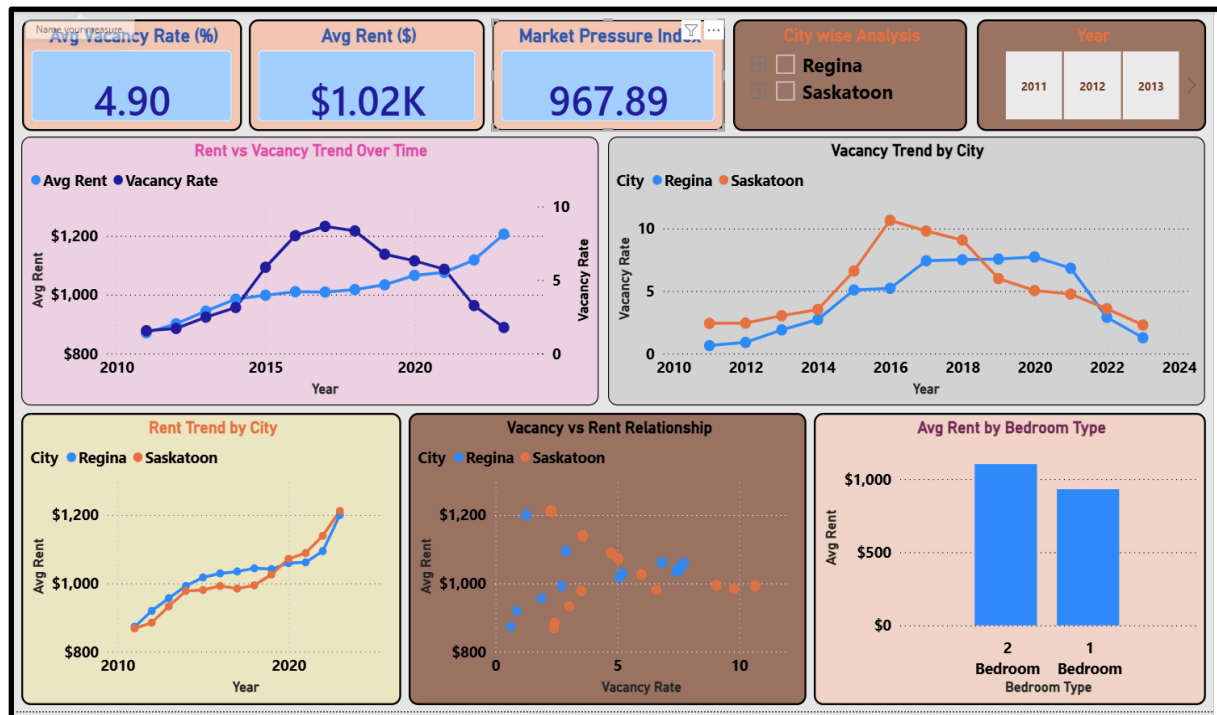


Figure 12 - Rental Market Pressure

This dashboard analyzes rental market pressure over the time for both cities

- **Rent vs Vacancy Trend Over Time**

This visual shows the relationship between rent and vacancy rate over time. From 2011 to around 2016–2017, vacancy rates increase, indicating more availability in the market, while rent continues to rise gradually. However, after this period, vacancy rates start declining, while rent continues to increase. This inverse movement suggests tightening market conditions, where reduced availability leads to higher rental prices.

- **Vacancy Trend by City**

The city-level comparison shows that Saskatoon initially experiences higher vacancy rates compared to Regina, peaking around 2016. However, in later years, vacancy rates decline in both cities. Regina shows a more stable pattern, while Saskatoon experiences sharper fluctuations, indicating more variability in housing availability.

- **Rent Trend by City**

This visual shows that rent increases steadily in both Regina and Saskatoon over time. Saskatoon shows slightly sharper growth in later years, aligning with earlier observations of increasing housing stress.

- **Vacancy vs Rent Relationship**

The scatter plot highlights the relationship between vacancy rate and rent. A general negative trend can be observed—higher vacancy rates are associated with relatively lower rents, while lower vacancy rates correspond to higher rents. This supports the idea that limited housing supply increases rental pressure.

- **Average Rent by Bedroom Type**

The comparison confirms that 2-bedroom units are consistently more expensive than 1-bedroom units. This reinforces earlier findings that larger households face greater affordability challenges.

Overall, this dashboard shows that declining vacancy rates and rising rent levels together contribute to increasing market pressure. As availability decreases, rental costs continue to rise, further worsening housing affordability across both cities.

5. Conclusion

This study developed a structured framework to analyze rental housing stress in Regina and Saskatoon over the period 2011 to 2023. By integrating rental data from CMHC and income data from Census and provincial growth reports, a consistent dataset was created to capture long-term trends in affordability.

The analysis introduced a composite Housing Stress Score, combining key indicators such as rent-to-income ratio, single-person vulnerability, income inequality, and tax impact. This allowed housing stress to be measured in a more comprehensive and interpretable way. The classification of stress into low, medium, and high categories further helped in understanding the severity of the problem across different zones and time periods.

The results indicate a clear increase in rental stress over time, especially after 2020. Rent growth has outpaced income growth, reducing affordability. Larger units (2-bedroom) show consistently higher financial pressure, and the gap between housing types continues to widen. Spatial analysis also shows that stress is unevenly distributed, with certain zones consistently experiencing higher pressure. Additionally, single-person households are found to be more vulnerable compared to general households.

Beyond analysis, this study provides practical value. The framework and dashboards can help local residents identify comparatively affordable zones, allowing them to make better housing decisions and manage their rent-to-income burden more effectively. It also supports policymakers in identifying high-stress zones, enabling targeted interventions such as rent regulation, housing support programs, or supply expansion. Further, urban planners can use these insights to understand spatial pressure patterns and plan more balanced housing development.

Overall, the study not only explains the problem but also provides a usable approach to understand and address housing stress in a structured and data-driven manner.

6. Suggestion for Future Work

1. Dynamic Weights and Thresholds

Instead of fixed weights and classification thresholds, the model can be improved by making them dynamic so that they adjust automatically based on changing data patterns and economic conditions.

2. Incorporate Additional Socio-economic Factors

Extend the model by including factors such as inflation, cost of living, employment rates, and migration trends to capture a more complete view of housing affordability.

3. Policy Impact Simulation

Introduce scenario-based analysis to evaluate the impact of policies like rent control, income subsidies, and housing supply changes on rental stress levels.

4. Advanced Forecasting Models

Develop predictive models to estimate future housing stress trends, allowing early identification of high-risk zones.

5. Cloud-based Deployment with Live Data Integration

Deploy the full pipeline on a cloud platform and connect it with live data sources to enable continuous monitoring and real-time updates.

6. Scalability to Larger Regions

Expand the framework beyond Regina and Saskatoon to cover more cities, surrounding regions, and eventually a province-wide or national-level analysis.

7. References

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